

Clinically-Enriched AutoML with Multi-Agent LLM Reporting for Biomedical Signal Classification: Design and a Critical Methodological Evaluation

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ABSTRACT

Accuracy alone is a poor summary of a biomedical classifier: a model that misses malignant cases can still look excellent on paper. We present the decision layer of *BioStatusIA*, an open clinical decision-support system, which combines (i) an automated machine-learning (AutoML) module that compares six classifiers under a clinically-enriched metric set—sensitivity, specificity, Matthews correlation coefficient (MCC), Cohen’s κ , expected calibration error (ECE) and a McNemar A/B test—with class balancing (SMOTE/ADASYN) and SHAP-based interpretability, and (ii) a multi-agent layer of five CrewAI crews driven by a locally-hosted large language model that turns the numerical output into a structured radiology-style report. Beyond describing the design, the central contribution of this paper is a *critical self-audit* of the methodology: we identify and quantify four concrete threats to validity—normalization computed before the train/test split, oversampling applied before cross-validation, an inadvertently disabled sensitivity floor in model selection, and single-split point estimates on small datasets—and we contrast optimistic synthetic results (AUC=1.0) with the honest operating point on real breast-ultrasound data (AUC $\approx$ 0.80, recall $\approx$ 0.45). We argue that publishing such an audit is itself good practice for reproducible medical AI.

Index Terms— AutoML, clinical machine learning, model calibration, class imbalance, SHAP, multi-agent systems, large language models, reproducibility, medical AI ethics.

1. INTRODUCTION

The temptation to report a single accuracy number is especially dangerous in medicine. On an imbalanced screening dataset a classifier that always predicts “benign” can achieve high accuracy while being clinically useless, because it never raises the alarm on the cases that matter. A trustworthy medical decision-support system therefore needs (a) a metric suite that exposes sensitivity, calibration and agreement, not just accuracy; (b) an explicit account of how the winning model was chosen; and (c) a transparent statement of the methodological choices that could bias the reported numbers.

This paper describes and critically evaluates the decision layer of *BioStatusIA*. The companion paper covers ingestion and feature extraction across four data families; here we take the extracted feature vectors as input and study two subsystems built on top of them. The first is an AutoML module that trains and compares six classifiers under a clinically-enriched protocol. The second is a multi-agent reporting layer in which specialized software agents, each backed by a large language model (LLM), transform the

numerical results into a human-readable clinical narrative under a strict ethical template.

Our contributions are threefold. First, we specify a clinically-enriched evaluation protocol and the model-selection rule that a screening context demands. Second, we describe a multi-agent architecture that separates quantitative analysis from natural-language reporting, keeping the LLM away from the numbers it must not invent. Third—and most importantly—we perform a *methodological self-audit*, naming and quantifying the biases in our own pipeline, so that the reported results can be interpreted correctly and corrected in future work.

2. RELATED WORK

Automated machine learning seeks to remove manual trial-and-error from model selection [1]. In clinical settings the emphasis shifts from raw accuracy toward metrics that reflect diagnostic risk: sensitivity and specificity, the Matthews correlation coefficient—shown to be more informative than F1 and accuracy on imbalanced data [2]—and Cohen’s κ for chance-corrected agreement [3]. Calibration, the agreement between predicted probabilities and observed frequencies, is frequently neglected yet critical when a probability is shown to a clinician [4]. Class imbalance is commonly addressed with synthetic oversampling such as SMOTE [5] and ADASYN [6], implemented here through imbalanced-learn [7]; a well-known pitfall is that oversampling must be performed *inside* each cross-validation fold to avoid optimistic bias [8]. Model comparison should be tested for significance, for which McNemar’s test is the standard on a shared test set [9]. Interpretability is provided by SHAP values [10]. Finally, LLM-based multi-agent frameworks such as CrewAI have made it practical to compose role-specialized agents; their use in medicine, however, raises well-documented hallucination and safety concerns [11]. Our work sits at the intersection of these threads and, unusually, foregrounds the failure modes rather than hiding them.

3. THE AUTOML DECISION LAYER

3.1. Model pool and protocol

Six classifiers compete on every labelled dataset: logistic regression, k -nearest neighbours, an RBF support-vector machine, a random forest, gradient boosting, and a multi-layer perceptron. Features are standardized, the data are split into an 80/20 stratified holdout, and each model is additionally assessed with stratified 5-fold cross-validation on the training partition. For every fold and for the final holdout we record accuracy, sensitivity (recall of the positive class), specificity, precision, F1, ROC-AUC, MCC, Cohen’s

κ , ECE over ten bins, inference latency and training time. The two leading models are compared with a continuity-corrected McNemar test that reports χ^2 , a p -value, and a significance flag at $\alpha = 0.05$.

3.2. Class balancing and interpretability

When the training partition is imbalanced and the minority class has enough neighbours, SMOTE (or ADASYN) resamples it; the procedure fails safe, returning the original data when resampling is not viable, and—by construction—is applied only to the training partition so that the held-out test set remains untouched. The winning model is explained with SHAP: a `TreeExplainer` for tree ensembles and a sampled `KernelExplainer` otherwise, yielding a global ranking of feature importance that feeds both the interface and the population report.

3.3. Model-selection rule

The intended rule reflects screening priorities: among models above a minimum sensitivity, pick the highest AUC, so that the system prefers catching disease over marginal overall accuracy. The winner is persisted to disk for single-case inference, together with its scaler, feature names and metrics.

4. MULTI-AGENT REPORTING LAYER

Numbers are necessary but not sufficient for clinicians, so BioStatusIA adds a narrative layer built from five CrewAI crews and eight role-specialized agents orchestrated sequentially and backed by a locally-hosted LLM (Ollama, `qwen2.5`). An imaging pipeline chains a PDI engineer, a technical analyst, a data scientist and a radiologist agent; parallel crews handle tabular data, physiological signals (F1) and medical images (F3/F4); a fifth, tool-free crew produces a focused single-case report. Crucially, agents communicate through the filesystem: each tool persists a complete JSON artifact and returns only a short textual summary to the LLM, so the language model composes prose *around* numbers it cannot alter. The interactive report is constrained to five mandatory sections—principal finding, severity 1–5, comparison to reference, immediate recommendation, and an ethical disclaimer—and the reporting agents are deliberately given no analytical tools, preventing them from fabricating measurements.

5. RESULTS

5.1. Real imaging: the honest operating point

Table 1 reports the six-model comparison on 780 breast-ultrasound images (570 benign, 210 malignant). The random forest is selected on AUC (0.800) with the best accuracy (0.814), but its recall of 0.452 is the headline clinical result: on genuinely imbalanced data the default image branch misses more than half of the malignant cases. This is exactly the kind of number that a raw-accuracy report would bury, and it directly motivates moving class balancing into the default path.

Table 1: Six-model comparison on real BUSI data (780 images). Winner in bold.

Model	Acc.	Prec.	Recall	F1	AUC
LogisticReg.	0.744	0.750	0.071	0.130	0.768
KNN	0.776	0.630	0.405	0.493	0.764
SVM	0.776	1.000	0.167	0.286	0.744
RandomForest	0.814	0.760	0.452	0.567	0.800
GradientBoosting	0.750	0.556	0.357	0.435	0.756
MLP	0.731	0.500	0.548	0.523	0.713

5.2. Small and synthetic data: optimistic scores

On a 100-image balanced mini-set the MLP reached AUC 0.94 with sensitivity and specificity of 0.90. On a 50-row tabular panel and on 14-example synthetic ECG/DICOM/NIfTI sets, several models reached AUC 1.0 with perfect sensitivity, specificity, MCC and κ . We report these numbers precisely because they are *not* evidence of clinical performance: perfect scores on datasets with only a handful of test examples reflect the ease of separating synthetic or tiny samples, amplified by the methodological factors analyzed next.

6. CRITICAL METHODOLOGICAL EVALUATION

This section is the core contribution. We audited the implementation and identified four threats to validity, ordered by clinical impact.

6.1. T1 — Normalization leakage

The feature scaler is fit on the *entire* dataset before the train/test split, so test-set statistics influence the normalization of the training data. This is a textbook form of data leakage that inflates reported performance. The correct remedy is to fit the scaler on the training partition only—ideally by wrapping the scaler and estimator in a single pipeline so the fit is confined to each split.

6.2. T2 — Oversampling before cross-validation

SMOTE is applied to the training partition, after which stratified folds are drawn from the already-oversampled set. Synthetic points can then be generated from neighbours that later land in a validation fold, leaking information across folds and optimistically biasing the cross-validated metrics [8]. Oversampling belongs *inside* each fold, e.g. via an imbalanced-learn pipeline evaluated with cross-validation.

6.3. T3 — Disabled clinical selection floor

The stated selection rule prefers models with sensitivity ≥ 0.8 ; in the implementation the guard is written so that the sensitivity condition is short-circuited and every model qualifies, reducing the rule to pure maximum-AUC. On BUSI this is why a model with recall 0.452 is selected. The fix is a one-line correction, but its clinical consequence—false negatives in cancer screening—is severe, which is precisely why the audit matters.

6.4. T4 — Single-split point estimates on small data

All headline test metrics come from one fixed 80/20 split. On the 50-row and 14-example sets the test partition contains a handful of cases, so a single misclassification swings AUC dramatically and the reported 1.0 values carry no confidence interval. Repeated or nested cross-validation with reported variance, and DeLong intervals on AUC, are needed before any performance claim.

6.5. Secondary observations

The McNemar baseline in code compares the winner against the first model rather than the second-best, diverging from the documented “two best models” intent; GLCM texture uses a single distance and orientation; and the LLM reporting layer is not validated for factual grounding—indeed the local model produced multi-hour timeouts in our environment, so reports were generated from the deterministic numeric artifacts. None of these invalidate the architecture, but each must be disclosed.

7. DISCUSSION

Two lessons emerge. First, a rich metric suite is only as trustworthy as the protocol that produces it: MCC, κ and ECE are meaningless if the split leaks or the selection rule is silently disabled. Second, separating the numeric engine from the LLM narrator is a sound safety pattern—the language model cannot alter measurements it never touches—but it does not by itself make the narrative correct, and an unvalidated 3-billion-parameter model must not be trusted for clinical wording. The remediation roadmap is concrete and inexpensive: move scaling and oversampling inside cross-validation, restore the sensitivity floor, replace single-split estimates with repeated CV and confidence intervals, and add a factual-grounding check to the reporting agents.

8. ETHICAL CONSIDERATIONS

BioStatusIA is a decision-support tool, not a diagnostic authority. A mandatory disclaimer that its output does not replace a licensed physician appears in every screen and every report, including as a required section of each LLM narrative. The self-audit in this paper is itself an ethical instrument: by publishing the ways in which our numbers can mislead, we reduce the risk that a reader over-trusts them. We consider the honest reporting of a recall of 0.452 more valuable than a polished but leaky 1.0.

9. CONCLUSION

We described the AutoML and multi-agent reporting layers of BioStatusIA and subjected them to a critical methodological audit. The system contributes a clinically-enriched evaluation protocol and a safety-oriented separation between numeric analysis and LLM reporting; the audit contributes something rarer—a transparent, quantified account of four validity threats in our own pipeline and their fixes. We advocate this audit-first posture as a norm for reproducible medical machine learning, where the cost of an

unexamined optimistic result is measured in missed diagnoses.

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